MILITARY SPECIFICATION
CIRCUIT CARD ASSEMBLY: 11746274
INTERFACE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of Specification MIL-C-48411 comprise the complete requirements for Interface.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-48411	Circuit Card Assemblies, General Specification for

DRAWINGS

11746274	Circuit Card Assembly, Interface

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item definition.- This assembly is a single ended printed circuit board consisting of op-amps, gates, relays, and a square rooter necessary to provide an interface between the computer section and the coefficient section of the STG.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 11746274.

3.2 Characteristics.

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3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power supply.- The requirements of 3.2 shall be met with the following supplies:

a. +4VDC
b. +5VDC
c. +10VDC
d. +15VDC
e. -15VDC

3.2.1.2 Operational amplifier/square rooter network.- The input is applied to Pin 12 and outputs are yielded on Pins 5 & 8 shown in Table I.

3.2.1.3 Operational amplifier/high voltage - high current driver/relay network.- The inputs to the network are applied to Pins 40 & 2; outputs are yielded on Pins 26 & 19 shown in Table II.

3.2.1.4 High voltage - high current driver/relay/operational amplifier network #1.- The input is applied to Pin 4 and outputs are yielded on Pins 28 & 35 as shown in Table III.

3.2.1.5 High voltage - high current driver/relay/operational amplifier network #2.- The input to the network is applied at Pin 6 and outputs are yielded on Pins 13 & 18 as shown in Table IV.

3.2.1.6 High voltage - high current driver/relay/operational amplifier network #3.- The input is applied to Pin 6 and the outputs are yielded on Pins 7 & 14 as shown in Table V.

3.2.2 Environmental.- The assembly shall meet the environmental requirements of Specification MIL-C-48411.

3.2.3 Physical characteristics.- The physical characteristics shall be as specified on Drawing 11746274 and the engineering parts list.

3.3 Design and construction.

3.3.1 Production drawings.- This assembly shall be fabricated in accordance with the requirements of Specification MIL-C-48411, Drawing 11746274 and associated lists.

3.3.2 Standard of manufacture.- The standard of manufacture shall be those specified in Specification MIL-C-48411.

3.3.3 Workmanship.- The workmanship shall be as specified in Specification MIL-C-48411.

3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in Specification MIL-C-48411.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in Specification MIL-C-48411.

4.1.2 Special test and examination.- The assembly shall undergo first article and counter testing in accordance with 4.1.2 of Specification MIL-C-48411.

4.2 Quality conformance inspection.- The assembly shall be inspected and tested in accordance with 4.2 of Specification MIL-C-48411.

4.2.1 Test equipment. (see Figure 1)

4.2.2.1 Power supply.- The power supplies shall be as follows:

a. +6VDC ± 1mVDC, 100μA or greater and a ripple of less than 10mVRMS.

b. +10VDC ± 1mVDC, 100μA or greater and a ripple of less than 10mVRMS.

c. +15VDC ± 1mV, 100μA or greater and a ripple of less than 10mVRMS.

d. +20VDC ± . 1VDC, 100μA or greater and a ripple of less than 10mVRMS.

e. -20VDC ± .1VDC, 100μA or greater and a ripple of less than 10 mVRMS.

4.2.1.2 Oscilloscope.- Rise time of scope plus associated measuring probe of 50 nano seconds or less, sweep period of 50 nano seconds per centimeter of greater, input resistance greater than 8 megohms, input capacitance less than 25 picafarads.

4.2.1.3 Dummy loads.- The dummy loads shall consist of nine 1/4 watt 10K ohm resistors, four of which shall have tolerances of .1%, the other 5%.

4.2.1.4 Digital voltmeter.- The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input impedance of 10 megohms or greater on all voltage scales.

4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions for the purpose of these tests shall be defined as:

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a. Temperature between 60°F and 90°F.

b. Barometric pressure between 28 and 32 inches of mercury.

c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power supply.- All requirements shall be met with the following power supplies:

a. +4VDC ± 1mVDC.

b. +5VDC ± 1mVDC.

c. +10VDC ± 1mVDC.

d. +15VDC ± .1VDC.

e. -15VDC ± .1VDC.

4.2.2.2 Digital output.- The "0" or false output level shall be between 0.00 and 0.40 VDC. The "1" or true output level shall be between +2.40 and +5.00 VDC.

4.2.2.3 Test set-up.- The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

4.2.2.3.1 Operational amplifier/square rooter network test.- When the stated input is applied to Pin 12, output Pins 5 & 8 shall be as shown in Table I.

TABLE I

INPUT (VDC)	OUTPUTS (1) (VDC)	
PIN 12	PIN 5	PIN 8
0V	10V ± .1V	3.378V ± .1V
10V ± 1mV	0V ± .1V	Not applicable
4 ± 1mV	7.745 ± .1V	Not applicable

(1) Outputs shall be within ± .2V of stated values over extreme temperature range.

4.2.2.3.2 Operational amplifier/high voltage - high driver/relay network test.- When the stated inputs are applied to input Pins 40 & 2, output Pins 26 & 19 shall function as shown in Table II.

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TABLE II

INPUTS (VDC)		OUTPUTS (VDC)	
PIN 40	PIN 2	PIN 26	PIN 19
10V ± 1mV	Logic 0	-10V ± .1V	-6.83 ± .1V
5V ± 1mV	Logic 0	- 5V ± .1V	
	Logic 1	10V ± 1mV	

4.2.2.3.3 High voltage - high current driver/relay/operational amplifier network #1 test.- When the stated input is applied to Pin 4, output Pins 28 & 35 shall function as shown in Table III.

TABLE III

INPUT	OUTPUTS (VDC)	
PIN 4	PIN 28	PIN 35
Logic 0	0V ± 10mV	0V(1) ± 20mV
Logic 1	1.94V ± .1V	1.94V(1) ± .1V

(1) Shall rise from 0V ± 20 mV to 1.94V ± .1V within five sec. ± one sec.

4.2.2.3.4 High voltage - high current driver/relay/operational amplifier network #2 test.- When the stated input is applied to Pin 6, output Pins 13 & 18 shall be as shown in Table IV.

TABLE IV

INPUT	OUTPUTS (VDC)	
PIN 6	PIN 13	PIN 18
Logic 0	0V ± 20mV	0V ± 20mV
Logic 1	3.68V(1) ± .1V	10.0(2)V ± .1V

(1) Shall rise from 0V ± 20mV to 3.68V ± .1V within 10 sec. ± 2 sec.

(2) Shall rise from 0V ± 20 mV to 10.0V ± .1V within 5 sec. ± 1 sec.

4.2.2.3.5 High voltage - high current driver/relay/operation amplifier network #3.- When the stated input is applied to Pin 6, output Pins 7 & 14 shall be as shown in Table V.

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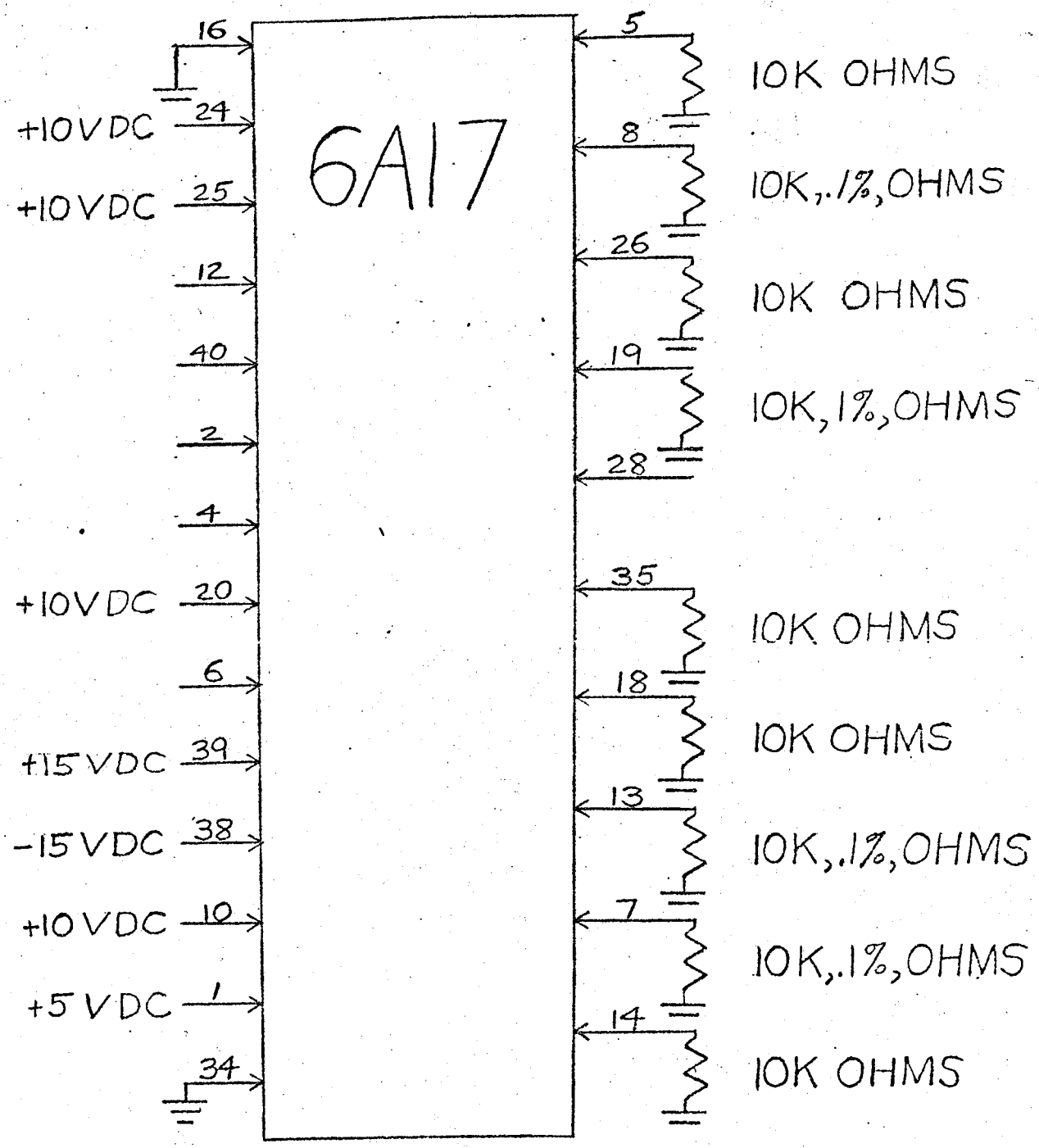


FIGURE 1

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TABLE V

INPUT	OUTPUTS (VDC)	
PIN 6	PIN 7	PIN 14
Logic 1	OV ± 20mV	OV ± 1mv
Logic 0	5.12(1) ± .2V	10V(2) ± .2V

(1) Shall rise from OV ± 20mV to 5.12V ± .2V within 10 sec. ± 2 sec.

(2) Shall rise from OV ± 1mV to 10V ± .2V within 10 sec. ± 2 sec.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11746274.

6. NOTES

6.1 The notes specified in the general Specification MIL-G-48411 are applicable to this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1240-A610

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